

Chapter Three

Security Techniques

Part – IV
Public-Key Cryptography
RSA

Software Security
(SE7432)

PUBLIC-KEY CRYPTOGRAPHY

- ✚ Traditional private/secret/single key cryptography uses one key shared by both sender and receiver
- ✚ If this key is disclosed communications are compromised
- ✚ Does not protect sender from receiver forging a message & claiming it sent by sender
- ✚ Public-key cryptography uses two keys – a public & a private key
- ✚ Public-key cryptography was designed by Whitfield Diffie & Martin Hellman at Stanford University in 1976
- ✚ The public-key cryptography involves the use of two keys:
 - ☞ Public-key, which may be known by anybody, and can be used to encrypt messages, and verify signatures
 - ☞ Private-key, known only to the recipient, used to decrypt messages, and sign (create) signatures

PUBLIC-KEY CHARACTERISTICS

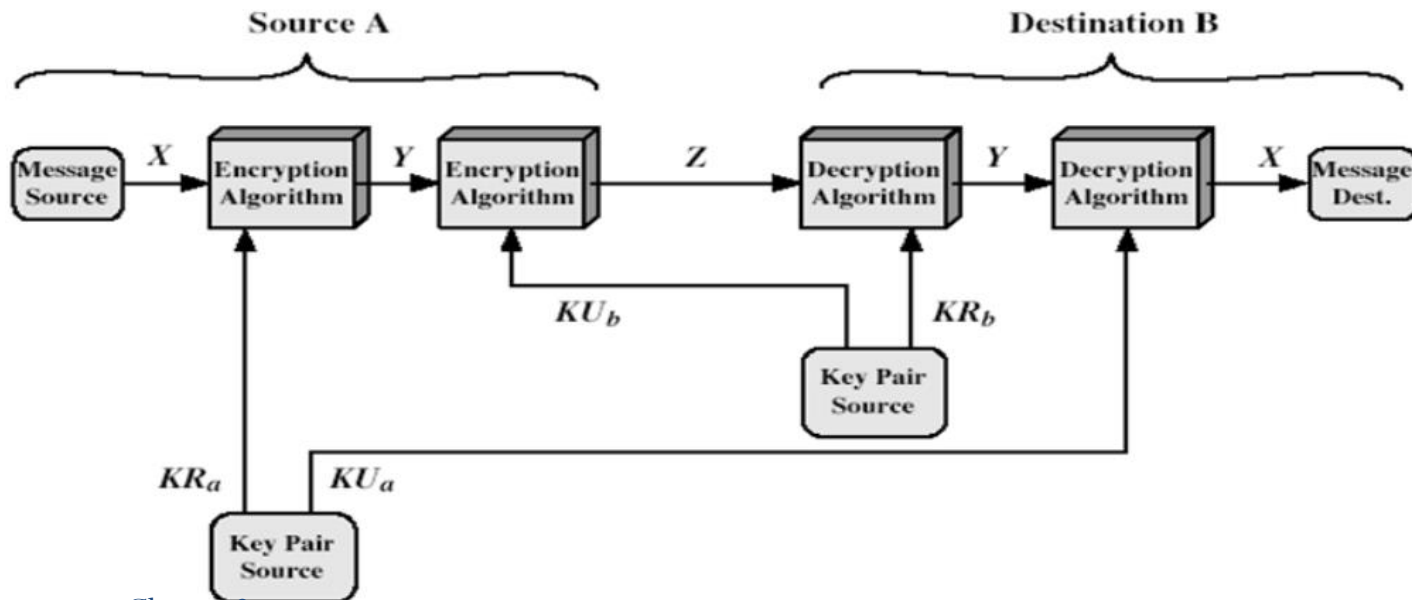
- ✧ Each user has two keys: a public key and a private key.
- ✧ The two keys operate as inverse, one key undoes the encryption provided by the other.
- ✧ A user can encrypt a message with a public key and the message can be revealed only with the corresponding private key. $P = D_{k_{\text{priv}}}(E_{k_{\text{pub}}}(P))$
- ✧ A user can encrypt a message with a private key and the message can be revealed only with the corresponding public key. $P = D_{k_{\text{pub}}}(E_{k_{\text{priv}}}(P))$
 - ☞ You can use either of the keys as private or public key for encryption and decryption
- ✧ Public-Key algorithms rely on two keys with the characteristics that it is:
 - ☞ Computationally infeasible to find decryption key knowing only algorithm & encryption key
 - ☞ Computationally easy to en/decrypt messages when the relevant (en/decrypt) key is known
 - ☞ either of the two related keys can be used for encryption, with the other used for decryption (in some schemes)

PUBLIC-KEY CRYPTOSYSTEMS

✚ Uses into 3 categories:

- ✚ Encryption/decryption (provide secrecy)
- ✚ Digital signatures (provide authentication)
- ✚ Key exchange (of session keys)

✚ Some algorithms are suitable for all uses, others are specific to one



PUBLIC-KEY CRYPTOSYSTEMS

- ✚ Like private key schemes brute force exhaustive search attack is always theoretically possible
- ✚ But keys used are too large (>512 bits)
- ✚ Security relies on a large enough difference in difficulty between easy (en/decrypt) and hard (cryptanalytic) problems
- ✚ More generally the hard problem is known, its just made too hard to do in practise
- ✚ Requires the use of very large numbers
- ✚ Hence is slow compared to private key schemes

COMPARING SECRET-KEY AND PUBLIC-KEY ENCRYPTION

	Secret key (Symmetric)	Public key (Asymmetric)
Number of Keys	One	Two
Protection of key	Must be kept secret	One key must be kept secret: the other can be freely exposed.
Best Uses	Cryptographic workhorse: secrecy and integrity of data- single characters to blocks of data, messages, files	Key exchange, authentication
Key Distribution	Must be out-of-band	Public key can be used to distribute other keys
Speed	Fast	Slow: typically, 10,000 times slower than secret key

RSA ENCRYPTION

- ✚ RSA cryptosystems, named after its inventors R. Rivest, S. Shamir and A. Adelman, is the most widely used public-key cryptosystems.
- ✚ It may be used to provide both secrecy and digital signatures and its security is based on the intractability of integer factorization problems.
- ✚ It is based on exponentiation in a finite (Galois) field over integers modulo a prime
exponentiation takes $O((\log n)^3)$ operations (easy)
- ✚ Uses large integers (eg. 1024 bits)
- ✚ Security due to cost of factoring large numbers: factorization takes $O(e \log n \log \log n)$ operations (hard)
- ✚ The RSA encryption algorithm combines results from number theory with the degree of difficulty in determining the prime factors of a given numbers.

RSA ENCRYPTION

- ✚ The two keys used in RSA, d and e , are used for decryption & encryption.
- ✚ They are actually interchangeable: Either can be chosen as the public but, having chosen one, you must keep the other one private. $P = E(D(P)) = D(E(P))$
- ✚ Any plaintext block P is encrypted as $P^e \bmod n$. Because the exponentiation is performed mod n , factoring P^e to uncover the encrypted plaintext is difficult.
- ✚ The decrypting key d is carefully chosen so that $(P^e)^d \bmod n = P$.

RSA KEY SETUP

- ✚ Each user generates a public/private key pair by selecting two large primes at random - p, q
- ✚ Computing their system modulus $N=p.q$, note $\phi(N)=(p-1)(q-1)$
- ✚ Selecting at random the encryption key e , where $1 < e < \phi(N)$, $\gcd(e, \phi(N))=1$
- ✚ Solve following equation to find decryption key d , $e.d=1 \bmod \phi(N)$ and $0 \leq d \leq N$
- ✚ Publish their public encryption key: $KU = \{e, N\}$
- ✚ Keep secret private decryption key: $KR = \{d, p, q\}$
- ✚ Users of RSA must:
 - ☞ Determine two primes at random - p, q
 - ☞ select either e or d and compute the other
 - ☞ Primes p, q must not be easily derived from modulus $N=p.q$, i.e. must be sufficiently large
 - ☞ Exponents e, d are inverses, so use Inverse algorithm to compute the other

RSA ENCRYPTION

✚ To encrypt a message M the sender:

✚ Obtains public key of recipient $K_U = \{e, N\}$

✚ Computes: $C = M^e \bmod N$, where $0 \leq M < N$

✚ To decrypt the cipher text C the owner:

✚ Uses their private key $K_R = \{d, p, q\}$

✚ Computes: $M = C^d \bmod N$

✚ Note that the message M must be smaller than the modulus N (block if needed)

RSA EXAMPLE

- ✚ Select primes: $p=17$ & $q=11$
- ✚ Compute $n = pq = 17 \times 11 = 187$
- ✚ Compute $\phi(n) = (p-1)(q-1) = 16 \times 10 = 160$
- ✚ Select e : $\gcd(e, 160) = 1$; choose $e=7$
- ✚ Determine d : $de = 1 \pmod{160}$ and $d < 160$ Value is $d=23$ since $23 \times 7 = 161 = 10 \times 160 + 1$
- ✚ Publish public key $K_U = \{7, 187\}$
- ✚ Keep secret private key $K_R = \{23, 17, 11\}$
- ✚ Sample RSA encryption/decryption is:
 - ✚ Given message $M = 88$ (nb. $88 < 187$)
 - ✚ Encryption: $C = 88^7 \pmod{187} = 11$
 - ✚ Decryption: $M = 11^{23} \pmod{187} = 88$

RSA SECURITY

- ✚ Three approaches to attacking RSA:
 - ☞ Brute force key search (infeasible given size of numbers)
 - ☞ Mathematical attacks (based on difficulty of computing $\phi(N)$, by factoring modulus N)
 - ☞ Timing attacks (on running of decryption)
- ✚ Compare and contrast: Substitution and Transposition, Stream and Block and Symmetric and Asymmetric encryption algorithms.
- ✚ Exercise crypt analysis attacks in example.
- ✚ Try to perform decryption mechanisms for all types of encryptions and apply cryptanalysis techniques to breach confidentiality of information